

8 Register mapping

The table given below provides a list of the 8/16-bit registers embedded in the device and the corresponding addresses.

Table 19. Registers address map

Name	Type	Register address		Default	Comment
		Hex	Binary		
RESERVED	-	00	00000000	-	Reserved
FUNC_CFG_ACCESS	r/w	01	00000001	00000000	Embedded functions configuration register
RESERVED	-	02	00000010	-	Reserved
RESERVED	-	03	00000011	-	Reserved
SENSOR_SYNC_TIME_FRAME	r/w	04	00000100	00000000	Sensor sync configuration register
SENSOR_SYNC_RES_RATIO	r/w	05	00000101	00000000	
FIFO_CTRL1	r/w	06	00000110	00000000	FIFO configuration registers
FIFO_CTRL2	r/w	07	00000111	00000000	
FIFO_CTRL3	r/w	08	00001000	00000000	
FIFO_CTRL4	r/w	09	00001001	00000000	
FIFO_CTRL5	r/w	0A	00001010	00000000	
DRDY_PULSE_CFG_G	r/w	0B	00001011	00000000	
RESERVED	-	0C	00001100	-	Reserved
INT1_CTRL	r/w	0D	00001101	00000000	INT1 pin control
INT2_CTRL	r/w	0E	00001110	00000000	INT2 pin control
WHO_AM_I	r	0F	00001111	01101010	Who I am ID
CTRL1_XL	r/w	10	00010000	00000000	Accelerometer and gyroscope control registers
CTRL2_G	r/w	11	00010001	00000000	
CTRL3_C	r/w	12	00010010	00000100	
CTRL4_C	r/w	13	00010011	00000000	
CTRL5_C	r/w	14	00010100	00000000	
CTRL6_C	r/w	15	00010101	00000000	
CTRL7_G	r/w	16	00010110	00000000	
CTRL8_XL	r/w	17	0001 0111	00000000	
CTRL9_XL	r/w	18	00011000	00000000	
CTRL10_C	r/w	19	00011001	00000000	

Table 19. Registers address map (continued)

Name	Type	Register address		Default	Comment
		Hex	Binary		
MASTER_CONFIG	r/w	1A	00011010	00000000	I ² C master configuration register
WAKE_UP_SRC	r	1B	00011011	output	Interrupt registers
TAP_SRC	r	1C	00011100	output	
D6D_SRC	r	1D	00011101	output	
STATUS_REG	r	1E	00011110	output	Status data register for user interface
RESERVED	-	1F	00011111	-	
OUT_TEMP_L	r	20	00100000	output	Temperature output data registers
OUT_TEMP_H	r	21	00100001	output	
OUTX_L_G	r	22	00100010	output	Gyroscope output registers for user interface
OUTX_H_G	r	23	00100011	output	
OUTY_L_G	r	24	00100100	output	
OUTY_H_G	r	25	00100101	output	
OUTZ_L_G	r	26	00100110	output	
OUTZ_H_G	r	27	00100111	output	
OUTX_L_XL	r	28	00101000	output	Accelerometer output registers
OUTX_H_XL	r	29	00101001	output	
OUTY_L_XL	r	2A	00101010	output	
OUTY_H_XL	r	2B	00101011	output	
OUTZ_L_XL	r	2C	00101100	output	
OUTZ_H_XL	r	2D	00101101	output	
SENSORHUB1_REG	r	2E	00101110	output	Sensor hub output registers
SENSORHUB2_REG	r	2F	00101111	output	
SENSORHUB3_REG	r	30	00110000	output	
SENSORHUB4_REG	r	31	00110001	output	
SENSORHUB5_REG	r	32	00110010	output	
SENSORHUB6_REG	r	33	00110011	output	
SENSORHUB7_REG	r	34	00110100	output	
SENSORHUB8_REG	r	35	00110101	output	
SENSORHUB9_REG	r	36	00110110	output	
SENSORHUB10_REG	r	37	00110111	output	
SENSORHUB11_REG	r	38	00111000	output	
SENSORHUB12_REG	r	39	00111001	output	

Table 19. Registers address map (continued)

Name	Type	Register address		Default	Comment
		Hex	Binary		
FIFO_STATUS1	r	3A	00111010	output	FIFO status registers
FIFO_STATUS2	r	3B	00111011	output	
FIFO_STATUS3	r	3C	00111100	output	
FIFO_STATUS4	r	3D	00111101	output	
FIFO_DATA_OUT_L	r	3E	00111110	output	FIFO data output registers
FIFO_DATA_OUT_H	r	3F	00111111	output	
TIMESTAMP0_REG	r	40	01000000	output	Timestamp output registers
TIMESTAMP1_REG	r	41	01000001	output	
TIMESTAMP2_REG	r/w	42	01000010	output	
RESERVED	-	43-48		-	Reserved
STEP_TIMESTAMP_L	r	49	0100 1001	output	Step counter timestamp registers
STEP_TIMESTAMP_H	r	4A	0100 1010	output	
STEP_COUNTER_L	r	4B	01001011	output	Step counter output registers
STEP_COUNTER_H	r	4C	01001100	output	
SENSORHUB13_REG	r	4D	01001101	output	Sensor hub output registers
SENSORHUB14_REG	r	4E	01001110	output	
SENSORHUB15_REG	r	4F	01001111	output	
SENSORHUB16_REG	r	50	01010000	output	
SENSORHUB17_REG	r	51	01010001	output	
SENSORHUB18_REG	r	52	01010010	output	
FUNC_SRC1	r	53	01010011	output	Interrupt registers
FUNC_SRC2	r	54	01010100	output	
WRIST_TILT_IA	r	55	01010101	output	Interrupt register
RESERVED	-	56-57		-	Reserved
TAP_CFG	r/w	58	01011000	00000000	Interrupt registers
TAP_THS_6D	r/w	59	01011001	00000000	
INT_DUR2	r/w	5A	01011010	00000000	
WAKE_UP_THS	r/w	5B	01011011	00000000	
WAKE_UP_DUR	r/w	5C	01011100	00000000	
FREE_FALL	r/w	5D	01011101	00000000	
MD1_CFG	r/w	5E	01011110	00000000	
MD2_CFG	r/w	5F	01011111	00000000	
MASTER_CMD_CODE	r/w	60	01100000	00000000	

Table 19. Registers address map (continued)

Name	Type	Register address		Default	Comment
		Hex	Binary		
SENS_SYNC_SPI_ERROR_CODE	r/w	61	0110 0001	00000000	
RESERVED	-	62-65		-	Reserved
OUT_MAG_RAW_X_L	r	66	01100110	output	External magnetometer raw data output registers
OUT_MAG_RAW_X_H	r	67	01100111	output	
OUT_MAG_RAW_Y_L	r	68	01101000	output	
OUT_MAG_RAW_Y_H	r	69	01101001	output	
OUT_MAG_RAW_Z_L	r	6A	01101010	output	
OUT_MAG_RAW_Z_H	r	6B	01101011	output	
RESERVED	-	6C-72		-	Reserved
X_OFS_USR	r/w	73	01110011	00000000	Accelerometer user offset correction
Y_OFS_USR	r/w	74	01110100	00000000	
Z_OFS_USR	r/w	75	01110101	00000000	
RESERVED	-	76-7F		-	Reserved